

Master's Thesis

A Study on Active Inference based Exploration in Model-Based Reinforcement Learning

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Declaration

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Abstract

Exploration remains a core challenge in reinforcement learning (RL), particularly in environments with sparse rewards or uncertain dynamics. While many approaches incorporate heuristic bonuses to encourage exploration, the framework of Active Inference offers a more principled alternative. Through the Expected Free Energy (EFE) objective, Active Inference decomposes behavior into both extrinsic reward maximization and epistemic exploration via information gain. In this work, we investigate how the state information gain component of EFE can be integrated into a model-based RL framework, specifically Dreamer. We propose an approach that approximates the divergence between posterior and prior latent beliefs using a classifier trained to distinguish between encoder and the dynamics model outputs. The classifier’s log-odds is then used as a pointwise intrinsic reward during imagined rollouts. We evaluate this method across several continuous control tasks from the DeepMind Control Suite and Gymnasium MuJoCo environments. While the classifier-based state information gain does not consistently outperform baseline Dreamer, our results offer insights into the sensitivity and potential of using theoretically motivated intrinsic signals to improve exploration in latent-space RL. This study contributes to bridging Active Inference with practical reinforcement learning, identifying key challenges and opportunities for future research.

Acknowledgments

First and foremost, I would like to thank...

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1 Introduction

Reinforcement Learning (RL) has seen significant progress in recent years, particularly through deep learning and model-based methods that enable agents to learn world models and plan actions in complex environments. However, effective exploration remains a central challenge in RL, especially in high-dimensional or sparse-reward settings. Agents must go beyond reward maximization to seek out informative experiences that improve their internal representations of the environment.

Active Inference, a framework originally developed within theoretical neuroscience, offers a principled perspective on this problem. Under the Active Inference paradigm, agents minimize a quantity called Expected Free Energy (EFE), which naturally decomposes into multiple intrinsic and extrinsic components: reward maximization, information gain about states, and control costs. This decomposition motivates behavior that balances exploitation and exploration—not through heuristics or added bonuses, but as a consequence of approximate Bayesian inference. Particularly relevant is the *state information gain* term in EFE, which quantifies how much an agent’s belief over latent states is expected to change upon observing future outcomes. This epistemic term provides a theoretically grounded signal for guiding exploration.

Despite its conceptual appeal, Active Inference has seen limited integration into mainstream deep RL algorithms. Most contemporary RL methods—including actor-critic and world model-based methods—optimize task-specific returns directly, often augmenting reward with extrinsic bonuses or hand-designed exploration strategies. One such example is Dreamer [1, 2], a model-based RL framework that learns a

latent dynamics model and performs policy optimization entirely in imagination. While Dreamer incorporates some elements reminiscent of variational inference (e.g., ELBO-based world model training), it lacks explicit treatment of the epistemic terms found in the EFE formulation. Notably, Dreamer does not utilize the state information gain between imagined posterior and prior beliefs, which limits its capacity for uncertainty-aware exploration.

In this work, we study how the Active Inference-inspired *state information gain* term can be integrated into Dreamer-style model-based RL. Our central question is: *Can epistemic uncertainty—formulated as the divergence between imagined posterior and prior beliefs—serve as a useful intrinsic reward in a Dreamer-like framework?*

To this end, we propose a novel method that approximates the state information gain using a discriminative classifier trained to distinguish between latent states generated from the posterior (encoder) and the prior (dynamics model). During imagination rollouts, the classifier’s log-odds output is used as a pointwise intrinsic reward, encouraging the agent to seek trajectories that would, in principle, yield informative observations.

We evaluate this approach across continuous control tasks from both the DeepMind Control Suite and Gymnasium MuJoCo environments, comparing it against a standard Dreamer baseline. While our results do not consistently outperform the baseline, this work contributes to the growing effort to bridge Active Inference and practical RL, and highlights both the promise and limitations of using theoretically grounded information-theoretic terms as exploration bonuses in deep model-based RL systems.

2 Related Work

We focus mainly on the relevant works for Model-Based methods in both Reinforcement Learning and Active Inference as they are more closely aligned to the framework of Active Inference.

2.1 Model-based Reinforcement Learning

Dreamer [1] is a prominent MBRL framework that learns a latent dynamics model to facilitate planning and policy learning entirely within a learned latent space, offering improved sample efficiency over model-free approaches. Dreamer leverages a Recurrent State-Space Model (RSSM) [3] to learn temporal dynamics and employs imagination-based rollouts for policy optimization using actor-critic methods. The original Dreamer (DreamerV1) demonstrated strong performance on pixel-based continuous control benchmarks [4]. Subsequent versions, such as DreamerV2 [5] and DreamerV3 [2], introduced improvements in stability, generalization, and scalability. DreamerV2 extended the method to discrete action spaces and showed promising results on Atari games, while DreamerV3 incorporated design changes to better handle both image-based and state-based observations, achieving strong performance across a wide range of benchmarks without environment-specific tuning.

While Dreamer learns the policy during imagination, its predecessor PlaNet [3] uses Cross Entropy Method (CEM) [6, 7] using the learned transition model. It introduces RSSM to learn the temporal latent dynamics model and uses it along with CEM

to select actions that maximize reward instead of an actor-critic. Both Dreamer variants and PlaNet are predominantly designed to work in a POMDP setting with pixel-based input and involve reconstruction of the image inputs during the training of RSSM.

Similarly to Dreamer, Aligned Latent Models (ALM) [8] also learn latent models and policy to jointly maximize the lower bound of the expected return. They also provide an intrinsic reward in the style of Expected Free Energy (EFE) in Active Inference during the actor-critic training, although it is confined to only Markov Decision Process setting and does not involve reconstructing observations.

2.2 Active Inference

Traditional AIF approaches were mostly limited to discrete spaces or short and finite-horizon POMDPs [9, 10], usually grid-world like environments. This limitation arises from the computational expense of evaluating each plan based on its EFE and selecting the next action from the plan with the lowest EFE [11, 12]. However, recent efforts have been made to extend AIF to POMDPs with continuous observation and/or action spaces. These efforts include approaches that focus on deriving the optimal distribution over actions (i.e., state-action policy) instead of plans or utilizing Monte Carlo Tree Search (MCTS) for plan selection. These advancements aim to address the computational challenges associated with generalizing AIF to continuous spaces.

Policy learning: In one of the very first works on deep Active Inference, [13] proposed sampling actions directly from a state-action policy rather than evaluating multiple planned trajectories. Their method minimized the Expected Free Energy (EFE) by approximating its gradient with respect to the policy parameters. A key challenge with this approach was the need to know how observations change in response to actions, which requires understanding the unknown transition dynamics

of the environment. To work around this limitation, [13] relied on an evolutionary genetic optimizer.

Building on this work, [14] developed a similar framework that incorporated the transition model explicitly. Their approach defined the optimal policy as a Softmax distribution over the EFE. Taking inspiration from deep Q-learning techniques [15, 16], they introduced a recursive bootstrapping method to approximate the EFE. However, it is important to note that their approach assumes that the current hidden state is fully known to the agent after inference and does not consider the agent’s belief state for action selection in the EFE and policy.

Preference Learning: Active Inference offers reward-free environments in their formulation and can achieve goal-directed behavior using other preferences. Contrastive Active Inference [17] focused on contrastive learning for the generative model and belief state in the environments with image-based observations. They also used a laplace distribution of the goal image as the preference distribution in the EFE without utilizing rewards. PPL [18] extends AIF to continuous observation spaces by showing that the minimum of the EFE follows a recursive form similar to the Bellman optimality equation in RL [19]. They gather expert data and learn an AIF based reward and view it as an inverse RL problem. However, their method is limited to discrete action spaces and only provides a deterministic policy. CRSP [20] extends this to pixel observation and continuous action spaces and gets a better performance than dreamerv3. They leverage few demonstrations of expert data to learn local state preferences which they use for efficiently solving sparse reward tasks

3 Background

In this section, we describe the formalisms of Model based RL, Active Inference and different types of Free Energies involved and how they are used. In the next chapter, we will discuss on how these both formalisms relate to each other

3.1 Reinforcement Learning

Reinforcement Learning (RL) is a framework for sequential decision making where an agent learns to interact with an environment to maximize cumulative reward. The environment is typically modeled as a Markov Decision Process(MDP) defined by the tuple $(\mathcal{S}, \mathcal{A}, P, r, \gamma)$ where $\mathcal{S}$ and $\mathcal{A}$ define the state and action spaces, $P(s'|s, a)$ defines the transition dynamics of the environment, $r(s, a)$ is the reward function and γ is the discount factor. At each time step, the agent observes a state s_t , selects an action a_t , receives a reward r_t , and transitions to a new state s_{t+1} . The goal of RL is to learn policies $\pi(a|s)$ that maximize the expected discounted reward:

$$\mathbb{E}_{\pi} \left[\sum_{t=0}^{\infty} \gamma^t r(s_t, a_t) \right]$$

3.1.1 Model-Based RL

Model-Based RL(MBRL) enhances the RL process by explicitly learning a model of the environment dynamics. Instead of relying solely on real environment interactions,

the agent learns an approximate transition model $\hat{P}(s'|s, a)$ and a reward model $\hat{r}(s, a)$ which it can use to simulate future outcomes, effectively reducing environment interactions. MBRL approaches are typically categorized into:

- Planning based methods, which optimize action sequences directly using the learned models [7]
- Latent imagination based methods which learn a compact latent dynamics model and use it to simulate imagined rollouts for training policies [8]

In this work, we focus on the latter class, where agents like DreamerV3 learn a latent space model and use it for actor-critic learning entirely in imagination. This structure makes them particularly amenable to augmentations such as intrinsic motivation or uncertainty-based exploration bonuses.

3.2 Active Inference

Active Inference is a process theory based on the Free Energy Principle(FEP) [21], a concept that recently emerged in the field of cognitive science. The FEP states [22] that the behavior of living systems can be understood as minimizing surprisal or maximizing model evidence, defined as the negative log marginal likelihood of the signals(o) they observe, $-\log P(o)$, under an embodied model of the environment $P(o)$. Given this premise, the role of perception is to make inference about the hidden causes of observed signals, the role of learning is to build models of the hidden causal structure and parameters in the environment, and the role of action is to intervene on the environment such that future signals are better predicted by the model. In this sense, active inference is a class of model-based reinforcement learning method. In contrast to the MBRL approach presented earlier, active inference makes a specific assumption that all parts of the perception-action process optimize a notion of free energy designed to overcome the intractability of calculating the log marginal

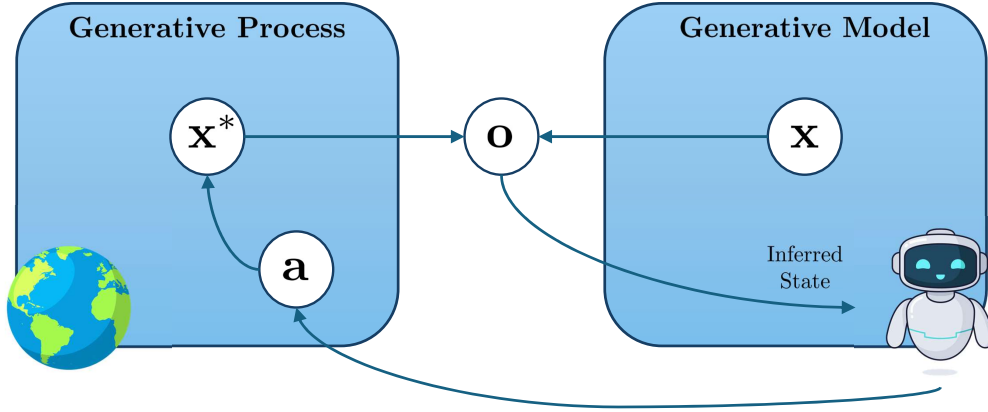


Figure 1: In Active Inference, the agent maintains a Generative Model of the world which sufficiently tries to explain the observations caused by the true Generative Process

likelihood. The attempt to unify perception and action gives active inference a special epistemic status compared to the traditional notion of MBRL.

3.2.1 Variational Free Energy

The Variational Free Energy (VFE) is a core quantity in variational inference [23] and constitutes a tractable bound on both the log model evidence and the KL divergence between prior and posterior. Let us say the agent exists in a partially observed world and receives some observation o from the environment and we wish to infer the hidden state of the world x . Meaning we would want to compute the posterior $P(x|o)$. While we do not know this posterior directly, we do possess the generative

model of the world which maps the hidden states to observations

$$P(o, x) = P(o|x)P(x)$$

FEP mandates that agents should minimise surprise

$$\min -\log P(o) \tag{3.1}$$

Marginalizing the hidden state x , we have

$$-\log P(o) = -\log \int P(o, x) dx \tag{3.2}$$

However $p(o, x)$ can be an arbitrarily complex distribution for which the integral above might not have an analytical solution. We tackle this by introducing a variational posterior $Q_\phi(x|o)$ parametrized by ϕ which we define and have full control over. Eq. 3.2 now becomes:

$$-\log P(o) = -\log \int P(o, x) dx \tag{3.3}$$

$$= -\log \int \frac{Q_\phi(x|o)}{Q_\phi(x|o)} P(o, x) dx \tag{3.4}$$

Using Jensen's inequality,

$$\leq - \int Q_\phi(x|o) \log \frac{P(o, x)}{Q_\phi(x|o)} dx \tag{3.5}$$

$$\leq \mathbb{E}_{Q_\phi(x|o)} [\log Q_\phi(x|o) - \log P(o, x)] \tag{3.6}$$

$$\leq \mathcal{F}(o, \phi) \tag{3.7}$$

where,

$$\mathcal{F}(o, \phi) := \mathbb{E}_{Q_\phi(x|o)} [\log Q_\phi(x|o) - \log P(o, x)] \tag{3.8}$$

$$= \mathbf{D}_{KL}[Q_\phi(x|o) || P(o, x)] \tag{3.9}$$

The quantity $\mathcal{F}(o, \phi)$ is defined as the Variational Free Energy(VFE) which serves as an upper bound on the surprise(or log model evidence). Depending on how the generative model $P(o, x)$ is decomposed, VFE takes two forms,

$$\begin{aligned}
\mathcal{F}(o, \phi) &= \mathbb{E}_{Q_\phi(x|o)}[\log Q_\phi(x|o) - \log P(o, x)] \\
&= \mathbb{E}_{Q_\phi(x|o)} \left[\log \frac{Q_\phi(x|o)}{P(x|o)P(o)} \right] \\
&= \mathbb{E}_{Q_\phi(x|o)} \left[\log \frac{Q_\phi(x|o)}{P(x|o)} \right] - \mathbb{E}_{Q_\phi(x|o)}[\log P(o)] \\
&= \underbrace{\mathbf{D}_{KL}[Q_\phi(x|o) || P(x|o)]}_{\text{Approx vs True Posterior}} - \underbrace{\log P(o)}_{\text{log evidence}} \tag{3.10}
\end{aligned}$$

$$\tag{3.11}$$

Since $\mathbf{D}_{KL}(\cdot || \cdot) \geq 0$, $\mathcal{F}(o, \phi) \geq -\log P(o)$. This decomposition shows that VFE acts as an upper bound for the surprise. Minimizing VFE with respect to the parameters of the Variational Distribution minimizes the KL Divergence and makes $Q_\phi(x_t|o_t)$ a good approximation of the true posterior while also making VFE a better estimate for the surprise.

While 3.10 shows theoretical properties of VFE, it does not provide a straight forward way to compute it. Decomposing the generative model in 3.8 the other way gives us

$$\begin{aligned}
\mathcal{F}(o, \phi) &= \mathbb{E}_{Q_\phi(x|o)}[\log Q_\phi(x|o) - \log P(o, x)] \\
&= \mathbb{E}_{Q_\phi(x|o)} \left[\log \frac{Q_\phi(x|o)}{P(o|x)P(x)} \right] \\
&= \mathbb{E}_{Q_\phi(x|o)} \left[\log \frac{Q_\phi(x|o)}{P(x)} \right] - \mathbb{E}_{Q_\phi(x|o)}[\log P(o|x)] \\
&= \underbrace{\mathbf{D}_{KL}[Q_\phi(x|o) || P(x)]}_{\text{Complexity}} - \underbrace{\mathbb{E}_{Q_\phi(x|o)}[\log P(o|x)]}_{\text{Accuracy}} \tag{3.12}
\end{aligned}$$

This decomposition is typically used in practice for VFE computation. This is also the ELBO loss typically used training Variational Auto Encoders(VAE)[24, 25] where

all the distributions involved in eq.3.12 would be gaussians.

Maximizing accuracy ensures observations are as likely as possible under the states x_t predicted by the variational posterior while keeping the posterior as close to the prior minimizing the complexity term. Essentially, it tries to maximize accuracy with the complexity term acting as a regularizer reducing the risk of overfitting to specific observations.

3.2.2 Expected Free Energy

Above section describes how the parameters of the agent’s generative model and variational posterior to minimize VFE. It is a passive way to accommodate the incoming observations into the agent’s models to minimize surprise Equation (3.1). Although the agent cannot control the observations, it can act on the environment to change the state and produce observations that minimize the surprise making the inference become active!(lending the name Active Inference).

Formally, while the above section allows agent to perform inference at the current time given observations, we can extend the formalism to allow for inference over actions or policies in the future. To implement this extension, we need a variational objective function that can be minimized based on future states and policies. This allows us to reframe adaptive action selection as a variational inference problem. This approach requires extending the current formalism in two key ways.

First, the generative model is augmented to include action a_τ and policies which are sequences of actions $\pi = [a_1, a_2...a_T]$. In order to transform action selection into an inference problem, policies are treated as an inferred distribution $Q(\pi)$ which is optimized to meet the agent’s goals. The second extension involves translating agent goals into a probabilistic framework. In active inference, goals are represented as a desired distribution over future observations $\tilde{P}(o_{\tau:T})$. The tilde notation indicates

that only the agent’s subjective distribution is biased, not the random variables themselves.

This goal distribution is incorporated into a biased generative world model:

$$\tilde{P}(o_\tau, x_\tau) \approx \tilde{P}(o_\tau)Q(x_\tau|o_\tau)$$

Here, we assume the true posterior can be well approximated by the variational posterior:

$$P(x_\tau|o_\tau) \approx Q(x_\tau|o_\tau)$$

indicating successful variational inference.

Active inference then derives a variational policy distribution $Q(\pi)$ that maximizes evidence for this biased model. This approach inverts traditional action selection. Rather than asking "What actions will achieve my goal?", the active inference agent asks "If my goals were achieved, what would have been the most probable actions I took?"

This motivates the definition of a variational objective $\mathcal{G}_\pi$, which quantifies how well a candidate policy π explains the biased generative model implied by the agent’s goals. Specifically, it measures the (negative) evidence lower bound (ELBO) or variational free energy under the goal-directed generative distribution $\tilde{P}$. Minimizing $\mathcal{G}_\pi$ encourages the agent to select policies that would render its desired future observations probable under its own generative model.

$$\mathcal{G} = \mathbb{E}_{Q(o_{t:T}, x_{t:T}, \pi)}[\log Q(x_{t:T}, \pi) - \log \tilde{p}(o_{t:T}, x_{t:T})]$$

To make this objective more tractable, we choose the variational distribution $Q(x_{\tau:T}, \pi)$ such that the hidden states of different time steps are independent using the temporal

mean-field approximation.

$$Q(x_{t:T}, \pi) \approx Q(\pi) \prod_t^T Q(x_\tau)$$

$$\tilde{P}(o_{t:T}, x_{t:T}) \approx \prod_t^T \tilde{P}(o_\tau) Q(x_\tau | o_\tau)$$

Inferring $Q(\pi)$ now becomes rather straight forward

$$\begin{aligned} \mathcal{G} &= \mathbb{E}_{Q(o_{t:T}, x_{t:T}, \pi)} [\log Q(x_{t:T}, \pi) - \log \tilde{p}(o_{t:T}, x_{t:T})] \\ &= \mathbb{E}_{Q(o_{t:T}, x_{t:T} | \pi) Q(\pi)} [\log Q(x_{t:T} | \pi) + \log Q(\pi) - \log \tilde{p}(o_{t:T}, x_{t:T})] \\ &= \mathbb{E}_{Q(\pi)} [\log Q(\pi) - \mathbb{E}_{Q(o_{t:T}, x_{t:T} | \pi)} [\sum_t^T [\log Q(x_\tau) - \log \tilde{P}(o_\tau, x_\tau)]]] \quad (3.13) \\ &= \mathbf{D}_{KL}[Q(\pi) || e^{-\sum_t^T \mathcal{G}_\tau(\pi)}] \end{aligned}$$

, where $\mathcal{G}_\tau(\pi) = \mathbb{E}_{Q(o_\tau, x_\tau | \pi)} [\log Q(x_\tau) - \log \tilde{P}(o_\tau, x_\tau)]$ is defined as the Expected Free Energy (EFE) for a single time-step τ . the optimal variational policy distribution $Q^*(\pi)$ is

$$Q^*(\pi) = \sigma(\sum_t^T \mathcal{G}_\tau(\pi)) \quad (3.14)$$

This implies that for inferring the optimal policy distribution, we can minimize the sum of expected free energies (EFE) across future time steps. The inference process works by:

1. Using the generative model to simulate potential futures
2. Computing the EFE for each predicted future state
3. Selecting policies that minimize the total expected free energy

Due to temporal mean field assumptions, trajectories can be broken down into individual time steps. This allows us to simplify our analysis to consider only a single

time step τ .

EFE can be decomposed into many ways similar to how VFE was decomposed. One notable decomposition is

$$\begin{aligned}
\mathcal{G}_\tau(\pi) &= \mathbb{E}_{Q(o_\tau, x_\tau | \pi)} [\log Q(x_\tau) - \log \tilde{P}(o_\tau, x_\tau)] \\
&\approx \mathbb{E}_{Q(o_\tau, x_\tau | \pi)} [\log Q(x_\tau) - \log \tilde{P}(o_\tau) - \log Q(x_\tau | o_\tau)] \\
&\approx \underbrace{-\mathbb{E}_{Q(o_\tau)} \mathbf{D}_{KL}[Q(x_\tau | o_\tau) || Q(x_\tau | \pi)]}_{\text{Epistemic Value}} - \underbrace{\mathbb{E}_{Q(o_\tau, x_\tau | \pi)} [\log \tilde{p}(o_\tau)]}_{\text{Extrinsic Value}} \quad (3.15)
\end{aligned}$$

Here the EFE separates into an extrinsic goal-directed term and an intrinsic Information seeking term (Epistemic value). The extrinsic term requires agents to increase the likelihood of the desired observations, $\tilde{p}(o_\tau)$, based on their beliefs about the future. This term guides the agent to act in a way that maximizes the probability of its desired outcomes occurring in the future. It is referred to as the extrinsic value term, as it represents the component of the EFE that reflects the agent's preferences.

The first term represents the expected information gain, often referred to as the 'epistemic value,' as it measures the information acquired by reaching a particular state. Since the information gain is negative, minimizing the EFE inherently requires maximizing the expected information gain. This encourages the agent to maximize the divergence between its posterior and prior beliefs, prompting actions that provide the most information and reduce uncertainty. The integration of extrinsic and intrinsic value terms underpins active inference's principled approach to addressing the exploration-exploitation trade-off [21, 12].

The concept of maximizing expected information gain, or "Bayesian surprise" has been advocated in neuroscience [26, 27, 28] and frequently proposed in reinforcement learning [29, 30, 31] to encourage exploratory behavior. However, it is worth noting that in these previous works, information gain is often introduced as an ad-hoc addition to an existing objective function, justified primarily by its intuitive role in enhancing exploration. In contrast, within the Expected Free Energy (EFE)

framework, expected information gain emerges naturally, providing the formalism with a certain degree of theoretical elegance.

Amortized Inference

In amortized variational active inference, we parametrize the policy $Q(\pi)$ or $Q(a_t|s_t)$ to approximate the variational action posterior $Q^*(\pi)$ in Eq. 3.14. This replaces the need for per-episode optimization with a learned inference network that generalizes across states. To make this inference tractable, we amortize the computation of the EFE across time steps using a recursive structure similar to the Bellman backup in reinforcement learning. The variational objective for a policy $Q(\pi)$ from Equation (3.13) is

$$\begin{aligned}\mathcal{G}(\pi) &= \mathbb{E}_{Q(\pi)}[\log Q(\pi) - \mathbb{E}_{Q(o_{t:T}, x_{t:T}|\pi)}[\sum_t^T [\log Q(x_\tau) - \log \tilde{P}(o_\tau, x_\tau)]]] \\ &= -\mathbb{E}_{Q(o_{t:T}, x_{t:T}|\pi)}[\sum_t^T \mathcal{G}_\tau(\pi)] - \mathbb{E}_{Q(x_{t:T}|\pi)} \mathbf{D}_{\mathbf{KL}}(Q(\pi) || P(\pi))\end{aligned}\quad (3.16)$$

where

$$\begin{aligned}G(x_t, \pi) &= -\mathbb{E}_{Q(o_\tau)} \mathbf{D}_{KL}[Q(x_\tau|o_\tau) || Q(x_\tau|\pi)] - \mathbb{E}_{Q(o_\tau, x_\tau|\pi)}[\log \tilde{p}(o_\tau)] \\ &\quad - \mathbb{E}_{Q(o_{t+1:T}, x_{t+1:T}|\pi)}[\sum_{t+1}^T \mathcal{G}_\tau(\pi)]\end{aligned}\quad (3.17)$$

3.2.3 Perception-Action Cycle

Using Variational Free Energy (VFE) to guide perception and learning, and Expected Free Energy (EFE) to guide decision-making, we can establish a complete perception-action cycle. This cycle describes how an agent perceives and interacts with its environment. For example, consider a person exploring a dimly lit room. Initially, they perceive their surroundings, forming a guess about the environment. Based on

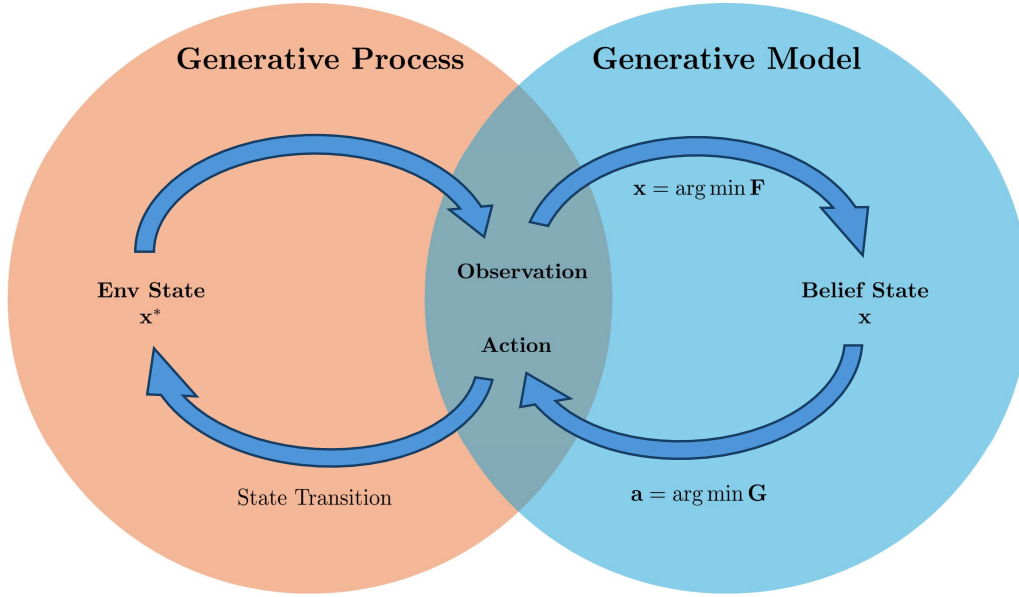


Figure 2: The perception-action cycle

this perception, they decide to light a candle (an action aimed at minimizing EFE). Lighting the candle alters the environment, changing what they perceive, and the cycle begins anew.

Figure 2 illustrates a simple example of this perception-action cycle. In essence, a change in the world state (x^*) produces an observation (o). Upon receiving this observation, the agent processes it, minimizing VFE to infer a posterior belief x about the world state. The agent then selects an action (a) to influence the world state (x) by minimizing EFE. This iterative process of perception, inference, and decision-making is known as the perception-action cycle.

We assume the environmental generative process follows a partially observable Markov decision process (POMDP), which incorporates two key concepts. First, partial ob-

servability means the agent cannot fully observe the hidden state of the environment, only acquiring partial information through observations. For instance, in dim lighting, one might only perceive vague outlines rather than clear details. Second, the Markov property implies that the current state depends only on the previous state, encapsulating all necessary information from past observations for decision-making. While the POMDP assumption may not hold in cases involving long-term memory, it effectively describes most scenarios. In control and reinforcement learning, POMDPs are commonly used as environmental models.

It is important to note that the agent’s generative model does not need to mirror the structure of the environmental generative process. The generative model’s goal is to minimize sensory prediction errors, not to replicate the environmental process perfectly. In fact, the generative model is often more expressive than the generative process. For example, when we walk and talk, our internal model generates observations based on a relatively simple (Markovian) environment.

4 Approach

In this work, we try to understand how the Information-Gain/Epistemic Value term arising from the Expected-Free Energy decomposition (Eq. 3.15), that provides an intrinsic motivation balancing exploration and exploitation, can be utilized in current MBRL frameworks. We mainly build upon the architecture of Dreamer.

4.1 Dreamer

The architecture of Dreamer [1] follows very closely to the structure of Active Inference based approaches. For the world model learning, following 3.12 for VFE,

$$\mathcal{F}(o, \phi) = \underbrace{\mathbf{D}_{KL}[Q_\phi(x|o)||P_\theta(x)]}_{\text{Complexity}} - \underbrace{\mathbb{E}_{Q_\phi(x|o)}[\log P_\theta(o|x)]}_{\text{Accuracy}}$$

and the Recurrent State Space Model(RSSM) training for dreamer is given by

$$\begin{aligned} \mathcal{L}_{RSSM} = & \mathbf{D}_{KL}[Q_\phi(x_t|x_{t-1}, a_{t-1}, o_t)||P_\phi(x_t|x_{t-1}, a_{t-1})] - \mathbb{E}_{Q_\phi(x|o)}[\log P_\phi(o_t|x_t)] \\ & - \mathbb{E}_{Q_\phi(x|o)}[\log P_\phi(r_t|x_t)], \end{aligned} \quad (4.1)$$

where $Q_\phi(x_t|x_{t-1}, a_{t-1}, o_t)$ is the variational posterior belief state after the observation o_t , and $P_\phi(x_t|x_{t-1}, a_{t-1})$ is the prior belief of the state. The KL divergence here is the Complexity term from VFE. The remaining two terms $\log P_\phi(o_t|x_t)$ and

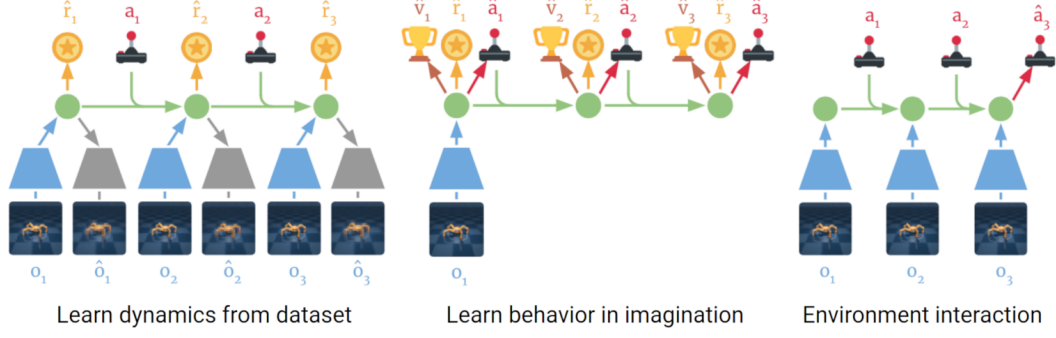


Figure 3: Components of Dreamerv1[1]

$\log P_\phi(r_t|x_t)$ are the likelihood terms for predicting observation and rewards. These both terms correspond to the accuracy term in VFE.

The RSSM [3] is a latent space dynamics model that is trained to predict the dynamics of the environment and predict the observations and rewards corresponding to the state. It is trained by collecting $\{(a_t, o_t, r_t)\}_{t=k}^{k+L}$ sequences from environment interaction and trained by Eq. 4.1. A key feature of RSSM, being a world model is, given an initial observation(or state x_t), and any sequence of actions $a_{0:t}$, the RSSM can be propagated forward to predict the latent states $x_{1:t+1}$ caused by the actions. This allows the agent to imagine state trajectories and corresponding rewards given actions.

For the Behavior learning, Dreamer uses actor-critic purely in the imagination conditioned on latent states. The action model $\pi_\theta(a_t|x_t)$ implements the policy and aims to predict actions that solve the imagination environment. The value model $v_\psi(x_t)$ estimates the expected imagined rewards that the action model achieves from each state.

The RSSM starting from an initial state s_t is rolled out over a horizon H using actions from policy $a \sim \pi_\theta(a_\tau|x_\tau)$ to collect imagination trajectory $\{(x_\tau, a_\tau, r_\tau)\}_{\tau=t}^{t+H}$. The target for the value model learning (Eq. 4.3) is the λ -weighted average of n-step return for different horizons $V_\lambda(x_\tau)$, while the objective for the policy (Eq. 4.2) is to predict actions that result in high value estimates.

The objectives to optimize are as following:

$$\max_{\theta} \mathbb{E}_{P_{\phi}, \pi_{\theta}} \left(\sum_{\tau=t}^{t+H} V_{\lambda}(x_{\tau}) \right), \quad (4.2)$$

$$\min_{\psi} \mathbb{E}_{P_{\phi}, \pi_{\theta}} \left(- \sum_{\tau=t}^{t+H} \ln v_{\psi}(V_{\lambda}(x_{\tau})) \right) \quad (4.3)$$

4.2 RL objectives from EFE decomposition

In the above section, we described how the RSSM world model training object appears directly from the VFE decomposition. In this section, we will analyze how the actor-critic objectives can be derived from EFE decompositions. From the amortized EFE decomposition (Eq. 3.16), we have:

$$\mathcal{G}(\pi) = \sum_t^T \mathcal{G}_{\tau}(\pi)$$

where

$$\mathcal{G}_{\tau}(\pi) \approx - \underbrace{\mathbb{E}_{Q(o_{\tau})} \mathbf{D}_{KL}[Q(x_{\tau}|o_{\tau})||Q(x_{\tau}|\pi)]}_{\text{Epistemic Value}} - \underbrace{\mathbb{E}_{Q(o_{\tau}, x_{\tau}|\pi)} [\log \tilde{p}(o_{\tau})]}_{\text{Extrinsic Value}} \quad (4.4)$$

$$- \underbrace{\mathbb{E}_{Q(x_{\tau})} \mathbf{D}_{KL}[Q(a_{\tau}|x_{\tau})||P(a_{\tau})]}_{\text{Control Cost}} \quad (4.5)$$

In RL, since we have no preference over other observations except on rewards, the preferred observation distribution can be $\tilde{p}(o_{\tau}) = \exp(r(\tau))$, essentially turning the extrinsic value term to reward maximization $\mathbb{E}_{Q(o_{\tau}, x_{\tau}|\pi)} [\log \tilde{p}(o_{\tau})] = \mathbb{E}_{Q(x_{\tau}, \pi)} [r(x_{\tau})]$.

Ignoring the state-information gain and the policy control cost from the EFE objective directly gives us the RL objective of reward maximization

$$\mathcal{G}(\pi) \approx \sum_t^T -\mathbb{E}_{Q(o_{\tau}, x_{\tau}|\pi)} [\log \tilde{p}(o_{\tau})] = - \sum_t^T \mathbb{E}_{Q(x_{\tau}, \pi)} r(x_{\tau})$$

This is directly the critic objective in dreamer and other RL methods, while for the actor objective is

$$\min_{\pi} \mathcal{G}(\pi)$$

For the later versions dreamer v2 and v3[5, 2], they also add the control cost with uniform action priors, reducing it to action entropy term

$$\mathbf{D}_{KL}[Q(a_{\tau}|x_{\tau})||P(a_{\tau})] = H[Q(a_{\tau}|x_{\tau})]$$

giving us the objective

$$\mathcal{G}(\pi) = \sum_t^T -\mathbb{E}_{Q(x_{\tau}, a_{\tau})}[r(x_{\tau}, a_{\tau}) - H[q(a_{\tau}|x_{\tau})]] \quad (4.6)$$

4.3 Understanding Informatation Gain

In the above decompositions of EFE to RL objectives, while the extrinsic value accounts for reward maximization and the control cost is accounted by the policy entropy, the epistemic value, also know as Information gain is often ignored in practice. This term, central to active inference, reflects the agent’s motivation to reduce uncertainty by seeking out informative experiences. To be precise, the information gain can be decomposed into two kinds from the decomposition of EFE.

$$\mathcal{G}_{\pi} = -\underbrace{\mathbb{E}_Q[\log P(x|o, \pi) - \log Q(x|\pi)]}_{\text{Information Gain(Hidden states)}} - \underbrace{\mathbb{E}_Q[\log P(\theta|x, o, \pi) - \log Q(\theta)]}_{\text{Information Gain(parameters)}} - \underbrace{\mathbb{E}_Q[\log \tilde{p}(o)]}_{\text{Extrinsic Vaue}}$$

Information Gain over hidden states (State IG) captures the expected change in the agent’s belief about the latent (hidden) state after receiving a new observation. The divergence between these two distributions reflects how informative the new observation is, relative to what the agent already believed. It captures epistemic uncertainty about the environment’s latent dynamics quantifying, “how surprising

or informative is this new observation, given what I thought would happen?”.

This form of information gain aligns closely with the classical notion of curiosity-driven exploration in RL [32, 33]. By visiting states that lead to large belief updates, the agent effectively explores novel or surprising regions of the environment — a behavior that can significantly accelerate learning in sparse or deceptive reward settings [34, 35].

Information Gain over parameters (Parameter IG) quantifies the expected change in the agent’s belief over the model parameters θ , given future observations and latent states. Conceptually, this measures the learning progress in the world model by measuring "how much an observation is expected to refine the agent’s understanding of the environment’s dynamics?"

In full Bayesian treatments of active inference, this component plays a key role in structure learning, allowing the agent to actively probe the world to improve its generative model. However, estimating this term in practice is extremely challenging:

It requires maintaining and updating a posterior distribution over the high-dimensional parameters of neural networks. Approximate solutions like variational Bayes [29], ensembles [36], Bayesian dropout add considerable complexity and may still be unreliable. As a result, modern implementations of active inference and information-seeking agents often neglect or approximate this term with heuristic proxies — such as learning progress, weight change, or model disagreement [37] — though these remain indirect and limited.

In this work, we specifically focus on incorporating the state information gain term into dreamer. There are several reasons for this choice:

- The belief over latent states is already explicitly modeled in Dreamer via the RSSM. We can therefore compute or estimate uncertainty over latent trajectories without modifying the model architecture or maintaining distributions over weights.

- Latent states are directly used by the actor-critic components for planning and decision-making. Hence, uncertainty in latent state estimation has an immediate influence on the policy and can be harnessed to improve exploration.
- Since Dreamer already separates inference (posterior) and prediction (prior) over latent states, the epistemic gap between them provides a natural signal to capture information gain. This structure aligns with the needs of approximating the first KL term in EFE decomposition.
- Despite its conceptual importance, state information gain is not used in most Dreamer variants. Integrating this signal may lead to improved exploration, better generalization, and more sample-efficient learning — especially in sparse reward tasks.

4.4 Estimating State Information Gain

We discussed how state information gain captures the agent’s epistemic uncertainty about latent states. It quantifies how much the observation at time t , denoted o_t , reduces uncertainty about the hidden state x_t , which is not directly observable. This reduction in uncertainty corresponds to an epistemic value term in the Expected Free Energy (EFE) decomposition and serves as a natural driver of curiosity-driven exploration.

Mathematically, the state information gain (State IG) is typically formalized as the KL divergence between the posterior and prior beliefs over the latent state.

$$\begin{aligned}
\text{State IG} &= \mathbb{E}_{Q(x_t, o_t | \pi)} [\log P(x_t | o_t, \pi) - \log Q(x_t | \pi)] \\
&\approx \mathbb{E}_{Q(x_t, o_t | \pi)} [\log Q(x_t | o_t, \pi) - \log Q(x_t | \pi)] \\
&= \mathbf{D}_{KL}(Q(x_t | o_t, \pi) || Q(x_t | \pi))
\end{aligned}$$

where $P(x_t|o_t)$ and $Q(x_t|o_t)$ are the true and the approximate posterior beliefs about state x_t given observation o_t , and $Q(x_t)$ is the prior belief about the state before observing

From a behavioral perspective, this KL divergence serves as a natural intrinsic reward. High divergence implies that the environment produced a surprising or novel observation relative to the model’s prior expectation, indicating epistemic uncertainty. By incorporating this signal into the policy’s reward structure, agents are encouraged to visit parts of the environment where their world model is uncertain or inaccurate, thereby improving exploration. The potential for using this term as intrinsic motivation has been noted in various active inference [32, 38] and curiosity-driven RL frameworks [33, 35, 34]

4.4.1 Reconstruction based State IG

Despite its elegance and theoretical appeal, directly integrating state information gain into Dreamer as an intrinsic reward is non-trivial. Dreamer’s Actor-Critic training happens in latent imagination, which uses the RSSM’s dynamics model to roll out in imagination without accessing the real observations. Since $Q_\phi(x_t|o_t)$ requires real observations, the state IG term cannot be computed in imagination rollouts.

A naive way to tackle this lack of observations in imagination, is to predict the observations and use them to calculate the imagined posterior. Using the observation reconstruction from Dreamer’s decoder and feeding them back into the encoder fig 4. Although this is similar to how Active inference’s method to use imagined observations, there is a subtle difference.

In Dreamer, imagined observations are generated by decoding latent states that are themselves sampled from a learned dynamics prior. When these decoded observations are fed again into the encoder, the resulting posteriors offer no new information beyond what the model already assumed. This is analogous to re-reading a summary

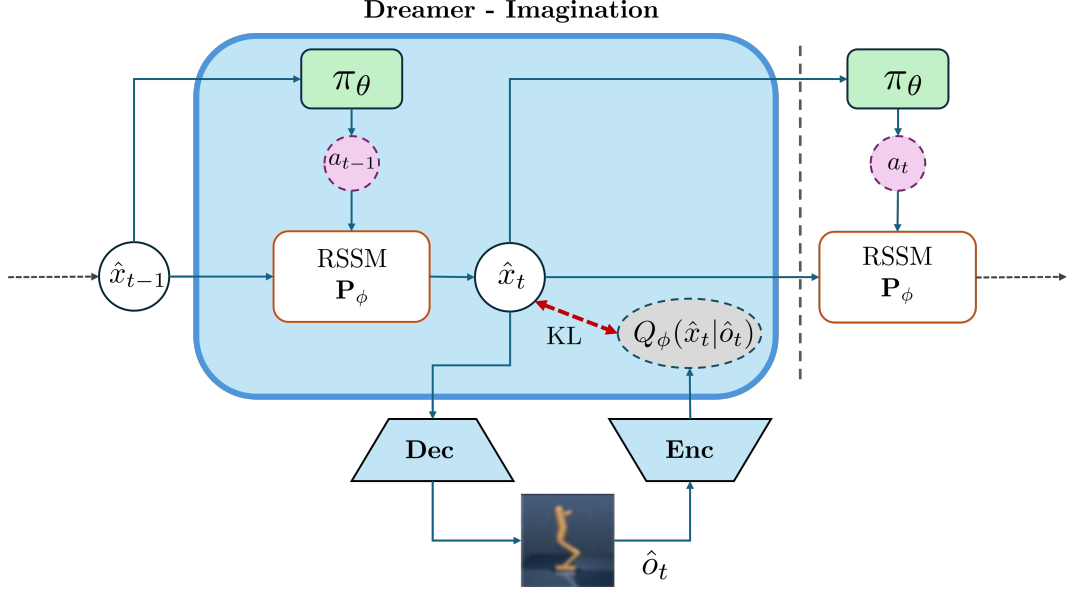


Figure 4: State IG between imagined posterior and prior during imagination rollout

you yourself wrote, there is no surprise, and thus no epistemic value. In contrast, Active Inference simulates belief updates for imagined observations that might contradict or enrich current beliefs, enabling the agent to select imagined outcomes that are expected to most reduce uncertainty. This proactive reasoning about information gain aids epistemic exploration in Active Inference which is missing in Dreamer without a mechanism to support belief updates over imagined outcomes.

In a hypothetical setup where the agent, during its imagination rollout, queries a parallel environment and observes real outcomes for the same actions, the information gain computed from posterior updates over these real observations would constitute a valid and accurate measure of epistemic value. However, this violates Dreamer’s core principle of imagination-based learning, as it requires interaction with the environment for every action considered in policy learning. This motivates the need for approximate methods to estimate epistemic value purely from imagination, such as our proposed classifier-based method

4.4.2 Approximating State IG with a Discriminator

As discussed in the previous sections, computing State IG directly for an architecture like dreamer is not feasible in a straightforward manner due to its requirement on accessing observations. This necessitates a way to compute State IG without access to real observations. A powerful and a general technique to approximate a KL Divergence between two distributions, especially when they are implicit distributions, is to use a Discriminative model, such as a binary classifier.

The core idea is that, if we train a binary classifier $D(x)$ to distinguish samples from two distributions $p(z)$ and $q(z)$ optimally, it outputs the probability

$$D^*(x) = \frac{q(z)}{p(z) + q(z)}$$

This result has been used extensively in the GAN literature [39], variational inference[40] and mutual information estimation[41]. Once the optimal discriminator is trained, the KL divergence between $p(x)$ and $q(x)$ can be written as

$$\frac{D(z)}{1 - D(z)} = \frac{q(z)}{p(z)} \implies D_{KL}(q(z)||p(z)) = \mathbb{E}_{x \sim q(z)}[\log \frac{D(z)}{1 - D(z)}] \quad (4.7)$$

4.5 Dreamer with classifier

In our case with dreamer, $q(z)$ is the posterior $Q(x_t|o_t)$ and $p(z)$ the prior $P(x_t|x_{t-1}, a_{t-1})$ and the samples are drawn from the posterior for training the discriminator. The discriminator is trained along with the RSSM world model fig. 5, although there will be no gradients flowing back into the RSSM. The classifier $C_\phi(x_t, a_{t-1}, x_{t-1}) \in [0, 1]$ is trained to distinguish samples from $x_t \sim Q_\phi(x_t|o_t)$ (label =1) and $\hat{x}_t \sim P_\phi(\hat{x}_t|x_{t-1}, a_{t-1})$ (label=0) predicting the probability that x_t is sampled from the encoder conditioned on the next observation starting from (x_{t-1}, a_{t-1}) pair. The

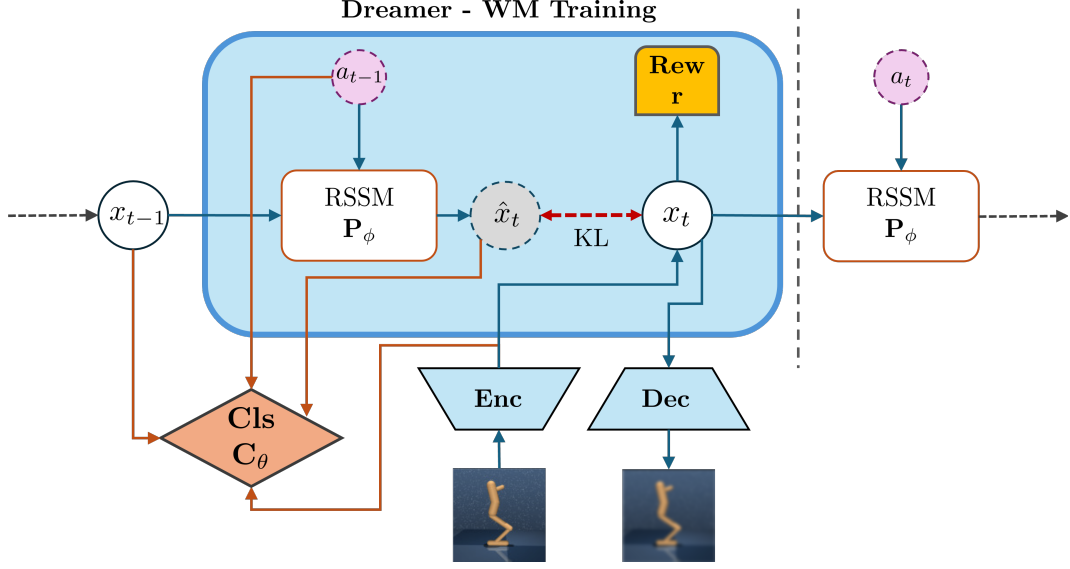


Figure 5: Classifier training during the model learning

classifier is trained via the standard cross entropy loss

$$\mathcal{L}_{c_\phi} = \log C_\phi(x_t, a_{t-1}, x_{t-1}) - \log (1 - C_\phi(\hat{x}_t, a_{t-1}, x_{t-1}))$$

4.5.1 Rollout Approximation

While the discriminator C_ϕ is trained to distinguish between posterior and prior samples during RSSM training, we aim to use it during imagination rollouts where only prior samples are available. This diverges slightly from eq? that the KL is computed as an expectation over posterior. However, once the discriminator is trained, we no longer require samples from both distributions to evaluate its output. The trained model effectively estimates the log-density ratio function

$$f(x_t) = \log \frac{Q(x_t)}{P(x_t)} \approx \log \frac{C_\phi(x_t)}{1 - C_\phi(x_t)}$$

This function can now be evaluated pointwise, even on prior samples from imagination. So, while the KL divergence is a global quantity (an expectation over the

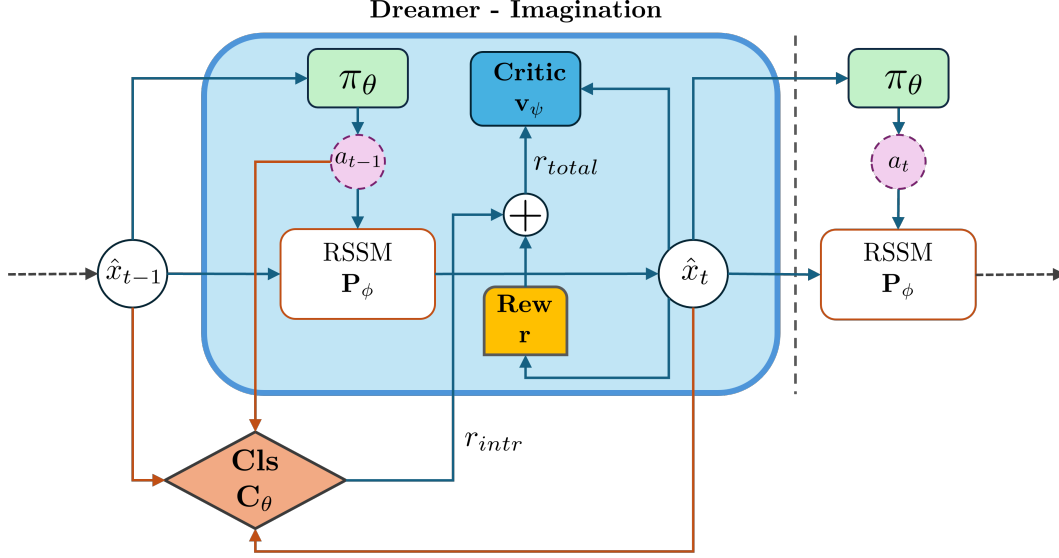


Figure 6: Classifier intrinsic reward during imagination

posterior), we instead use the pointwise log-density ratio $f(x_t)$ as a scalar intrinsic reward during rollout

$$r^{intr}(x_{t+1}) = \log \frac{C_\phi(x_{t+1}, x_t, a_t)}{1 - C_\phi(x_{t+1}, x_t, a_t)}, \text{ for } x_{t+1} \sim P_\phi(x_{t+1}|x_t, a_t) \quad (4.8)$$

This shift from expected KL divergence to pointwise log-ratio has precedent in several works that use classifier-based discriminators to generate scalar intrinsic rewards [8, 41, 42, 43].

Why Prior Samples? Even though we train the classifier using both posterior and prior samples, its purpose is to learn a decision boundary that distinguishes the two. After training, the classifier provides a valid estimate of the relative likelihood of a sample being from the posterior versus the prior. Evaluating it on prior samples lets us estimate, for each imagined state, how surprising (i.e., informative) it would be if an observation corresponding to it were acquired.

From Log-Odds to Information Gain To see how the classifier log-odds relate more directly, the mutual information between observation o_t and latent state x_t under a belief Q is given by:

$$I(o_t; x_t) = \mathbb{E}_{o_t \sim p(o_t)} [D_{\text{KL}}(Q(x_t \mid o_t) \parallel P(x_t))].$$

This is the classic state information gain quantifying how much we expect our belief to change after receiving observation o_t .

In practice, the KL divergence inside this expectation is approximated by the classifier. And since we cannot marginalize over observations in imagination, we settle for a pointwise estimate

$$\text{State IG at } x_{t+1} \approx \log \frac{Q(x_{t+1})}{P(x_{t+1})} \approx \log \frac{C_\phi(x_{t+1})}{1 - C_\phi(x_{t+1})}.$$

In this way, the log-odds term approximately reflects the amount of epistemic uncertainty or information gain associated with the imagined state. A high log-odds value implies that this state is one that, if actually observed, would result in a large update to our belief—making it a desirable target for exploration.

Therefore, even though we do not compute a full KL or marginal information gain, the classifier allows us to approximate a localized, per-state information gain signal during rollout.

4.5.2 Integrating to Dreamer

Finally, we integrate this approximate State IG via classifier as an intrinsic reward (Eq. 4.8) and provide it for the actor critic training during the imagination rollouts Fig. 6

$$r_{\text{total}} = r(x_t) + k r_{\text{intr}}(x_t)$$

where $r(x_t)$ is predicted by the reward head during rollouts.

We use the latest version of dreamer, the dreamerv3 as our baselines and build on top of it with the classifier. The updated bootstrapped λ -returns which serve as targets for the critic would then be

$$R_t^\lambda = r(x_t) + k r_{intr}(x_t) + \gamma c_t((1 - \lambda)v_\psi(x_t) + \lambda R_{t+1}^\lambda)$$

Although in theory (Eq. 4.5), the coefficient k for the intrinsic reward should be $k = 1$, due to the approximation, we use k between 0.01 and 0.1 in the experiments.

5 Experimental Evaluation

We call our variant C-Dreamer(Dreamer with classifier) with the intrinsic reward gain term. To evaluate the effect of the state IG term, we chose environments where there was larger scope for exploration. We provide the details of the environments in 5.1, the baselines used in 5.2, training settings in 5.3 and compare the results in 5.4.

5.1 Environments

To evaluate whether the state IG term in C-Dreamer contributes to improved performance, we selected environments where intrinsic motivation could influence exploration. Specifically, we chose the Cheetah and Hopper tasks from both the DeepMind Control Suite (DMC)[4] and Gymnasium Mujoco[44] benchmarks.

The Gymnasium environments typically offer higher rewards, on the order of 10^3 compared to DMC environments, where even high-performing agents usually attain rewards below a thousand. This contrast provides a useful spectrum for evaluating performance gains due to intrinsic exploration incentives.

The Cheetah environment (cheetah_run in DMC and HalfCheetah-v5 in Gymnasium) is relatively easier and allows a wide range of movement strategies to achieve higher rewards. In contrast, the Hopper environment (hopper_hop in DMC and Hopper-v5 in Gymnasium) presents a more challenging control problem, where agents

typically perform worse, making it a better testbed for evaluating whether state IG helps with exploration in harder tasks.

5.2 Baselines

To compare the performance of C-Dreamer, we evaluated it against two different types of baselines, Dreamer-v3 and Random Network Distillation (RND) agents. C-Dreamer was evaluated for different values of the classifier coefficient k . The architecture of dreamer-v3 is discussed quite elaborately in 4.1 and as it is the model on which C-Dreamer is built upon, it can provide a direct comparison if the intrinsic reward in C-Dreamer is having any effect. In the experiments, we refer to the vanilla Dreamer-v3 as $k = 0$. On the other hand, RND is a model-free agent which also uses intrinsic motivation for exploration, although it interacts with the environment and doesn't plan in latent space.

5.3 Training

Theoretically, the state information gain exists if there is a belief state being inferred from the observations, as there is no new information to gain if the state is fully observable. On that note, the experiments were predominantly carried out in POMDP settings where, the agents have access to only pixel observations in one setting, while in the other setting, the agents have access to only the joint positions and not the velocities. Many runs were tried with varying values of the classifier coefficient k for C-Dreamer, while all the other parameters remain identical to dreamerv3. The results were computed for 3 seeds per task per environment. Following the original implementation, the parameters of RND agents were unchanged and a frame stack of 4 frames history is given to the agent at every step. The agents were trained for 500k steps for low dimensional inputs and 1M steps for pixel inputs.

5.4 Results

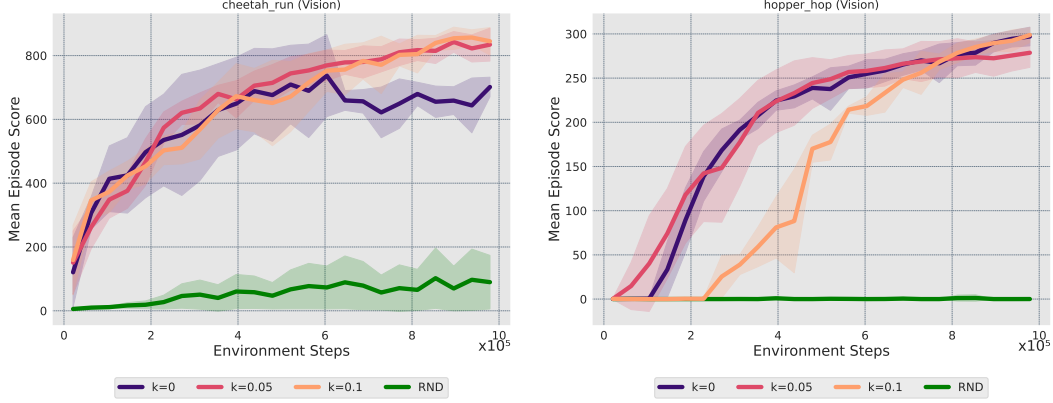


Figure 7: Results for all the methods on DMC vision benchmark

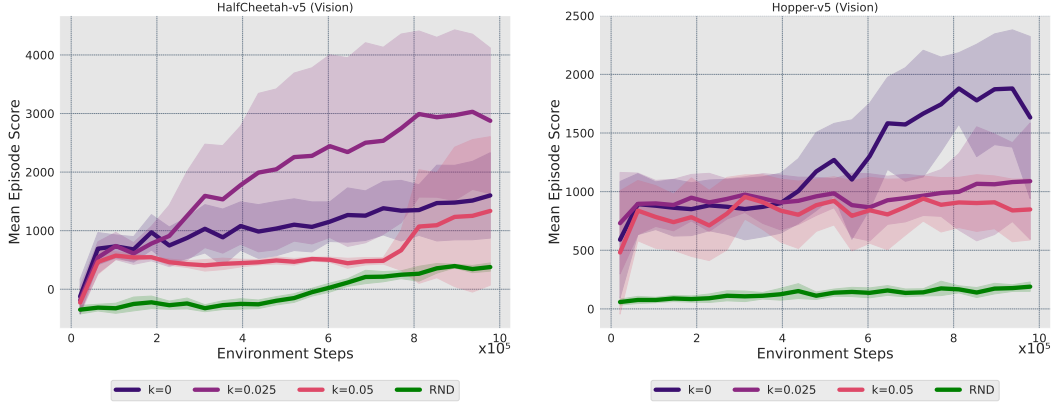


Figure 8: Results for all the methods on gymnasium vision benchmark

Figure 7 and Figure 8 show our experiment results on dmc and gymnasium vision benchmarks respectively. As discussed in 5.1, the gymnasium environments allow for more exploration in contrast to dmc, and this can be evidenced in C-dreamer’s superior performance on the HalfCheetah environment. However, this trend is not reflected in the hopper. Perhaps a deeper investigation could shed some light on this difference. In our partial observation experiments, we find exactly similar trend in the case of gymnasium cheetah while the dmc benchmarks for both the variants have been similar.

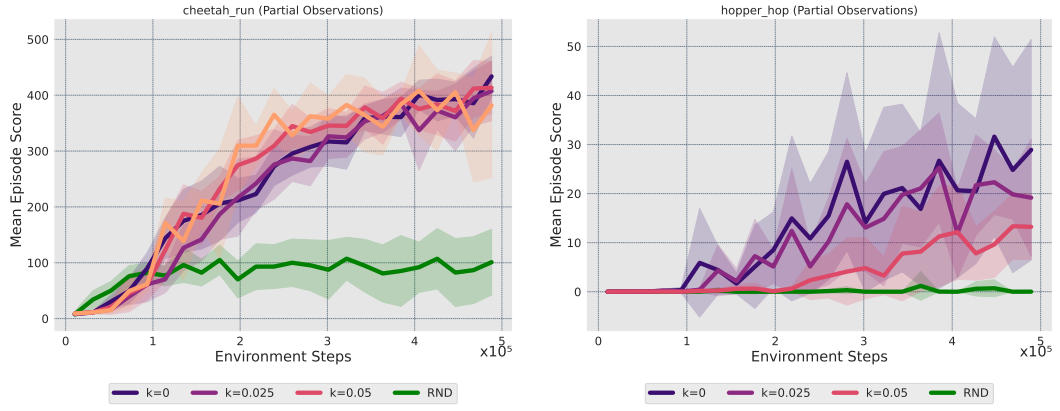


Figure 9: Results for all the methods on DMC partial observations

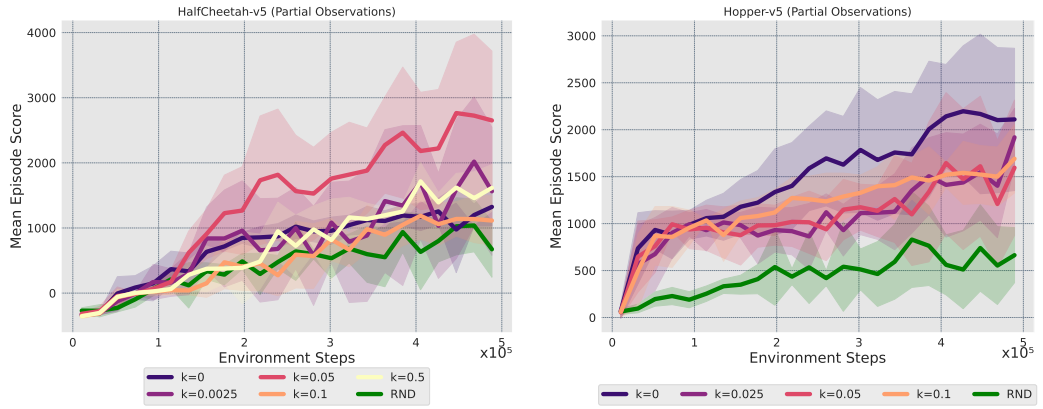


Figure 10: Results for all the methods on gymnasium partial observations benchmark

6 Conclusions and Future Work

In this work, we investigated the integration of Active Inference-inspired exploration—specifically, state information gain—into model-based reinforcement learning through the Dreamer framework. By training a discriminator to approximate the KL divergence between posterior and prior latent states, we proposed a practical method for computing intrinsic rewards without requiring real observations during imagination rollouts. This allowed us to incorporate epistemic uncertainty as a pointwise signal guiding exploratory behavior in Dreamer.

Our empirical evaluation across continuous control benchmarks from the DeepMind Control Suite and Gymnasium MuJoCo environments revealed a mixed picture. On Gymnasium’s Cheetah environments, the classifier-based Dreamer showed improved performance over the vanilla variant, both under visual and partially observed (state-only) settings. These results suggest that Active Inference-inspired objectives may be effective in environments where structured uncertainty can guide exploration. However, this trend did not consistently generalize to other environments such as Hopper or the DeepMind Control Suite tasks. In particular, high values of the intrinsic reward coefficient often destabilized learning, indicating sensitivity to hyperparameter tuning and a lack of robustness.

Overall, while our approach does not establish state information gain as a universally superior exploration strategy in model-based RL, it provides a concrete and scalable method for approximating such epistemic terms. The fact that improvements

emerged in a subset of settings is a promising sign that merits further investigation. Future work may explore alternative formulations of information gain, more expressive classifiers, or hybrid methods that better balance epistemic and extrinsic objectives. This study highlights both the promise and the practical challenges of bridging Active Inference principles with deep reinforcement learning.

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